D00310 / SAVED CURRENT ALARM		
Code	Alarm	Action / evidence
0	No alarm	No active saved fault
1	Safety not ready	M00001=0
2	Operator abort	STOP level or Auto→nonzero mode during20..110
101	Test transfer timeout	T0001
102	Clamp close timeout	T0002
103	Test response timeout	T0003
104	Invalid result	Matching DONE/ID but result outside1..3 →900
105	Clamp release timeout	T0004
106	Gate position timeout	T0005
107	Outfeed timeout	T0006
108	Reserved comms alarm	Reserved / not implemented
109	Gate home timeout	T0008
190	Unknown state	D00100 outside registered16 values

100ms BASE / CONFIRMED SETTINGS		
Timer	Function	Setting: counts /time
T0001	Transfer limit	D00200: 600 / 60s
T0002	Clamp close limit	D00201: 600 / 60s
T0003	Test response limit	D00202: 1200 / 120s
T0004	Clamp release limit	D00203: 600 / 60s
T0005	Gate arrival limit	D00204: 600 / 60s
T0006	Outfeed limit	D00205: 600 / 60s
T0008	Gate home limit	D00207: 600 / 60s

OUTPUT INTERLOCKS

All motion commands: safety AND NOT STOP AND alarm0.
Conveyor: (state30 OR (state90 AND M0006B)) AND M00076 AND NOT M01008.
Clamp command in40/50/60.
Gate command81..83/90 uses latched result D00102.
Reverse command remains OFF.
READY green / RUN yellow / ALARM red / buzzer until ACK.

Alarms are PLC-saved process faults. CIMON bridge errors and LabVIEW5xxx errors are separate namespaces; do not reuse them as D00310 codes.

108 / D00023 / T0007 are legacy-reserved contracts, not implemented MCU heartbeat logic. D00030 / T0009 are current PLC live supervision.

NO.	DATE	REVISION
R0	2026-10-22	FUNCTION RECORD

DWG. TITLE

SEMICON SORTER PROJECT
Final Test and Multi-Bin Sorting Cell

LS ALARM TIMER AND OUTPUT INTERLOCKS

DRAWN BY	DATE
J. H. KO	2026-10-14

DWG. NO
014

SHEET NO
1 OF 1